QUANT

Work & Interest

Assignment 02

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Work, Chain Rule & Interest Online Assignment -02

1. A can do a piece of work in 20 days. He worked for 5days. After this B did the remaining work in 3 days. How many days A and B will together take to finish the whole work?

1. 10 days 2. $\frac{10}{3}$ days 3. $\frac{5}{3}$ days 4. None of these

2. A man can do a piece of work in 5 days, but with the help of his son, he can do it in 3 days. In what time can the son do it alone?

1. $6\frac{1}{2}$ days 2. 7 days 3. $7\frac{1}{2}$ days 4. 8 days

3. A cistern is filled in 8 hours but takes two hours longer to fill because of a leak in its bottom. If the cistern is full, the leak will empty it in:

1. 40 hours 2. 20 hours 3. 10 hours 4. None of these

4. Two taps can separately fill a cistern in 10 min and 15 min. respectively and when waste pipe is open; they can together fill it in 18 minutes. The waste pipe can empty the full cistern in

1. 9 min 2. 12 min 3. 15 min 4. None of these

5. A and B can do a piece of work in 3 days, B and C in 4 days and A and C in 6 days. A, B and C can together complete the work in days?

1. $6\frac{2}{3}$ days 2. $2\frac{2}{3}$ days 3. $2\frac{1}{3}$ days 4. None of these

6. A can do a piece of work in 20 days. B can do the same work in 30 days. Working together, they can complete the work in

1. 12 days 2. 8 days 3. 10 days 4. 15 days

7. A tank is filled by a pipe A in 32 min. and pipe B in 36 min. When filled, it can be emptied by pipe C in 20 minute. If all the three pipes are opened simultaneously, 1/3rd of tank will be filled in

1. $55\frac{1}{13}$ min 2. $36\frac{12}{13}$ min 3. $35\frac{2}{13}$ min 4. None of these

8. A can do a work in 15 days and B in 20 days. If they work on it together for 4 days, then the fraction of the work that is left is:

1. $\frac{1}{4}$ 2. $\frac{1}{10}$ 3. $\frac{7}{15}$ 4. $\frac{8}{15}$

9. A tap can fill a cistern in 4 minutes. Another tap can empty it in 6 minutes. If both the taps are opened simultaneously, then the cistern will be filled in

1. 4 minutes 2. 6 minutes 3. 8 minutes 4. 12 minutes

10. A filling pipe can fill a tank in 20 min. There is a leak which can empty the tank in 60 min. When both these are working together, the tank would be filled in

1. 30 min 2. 36 min 3. 38 min 4. 40 min

Answer Key & Explanations

1.	2	A's one day work = $\frac{1}{20}$; A's = five day work = $\frac{5}{20} = \frac{1}{4}$, Remaining work = 1-$\frac{1}{4} = \frac{3}{4}$, B can do $\frac{3}{4}$ of work in 3 days, B can do 1 work in $3\times\frac{4}{3}$ = 4 days, B's one day work = $\frac{1}{4}$, A's one day work = $\frac{1}{20}$, (A+B)'s one day work $\frac{1}{4}+\frac{1}{20}=\frac{5+1}{20}=\frac{6}{20}=\frac{3}{10}$. So, A and B together can do the work in $\frac{10}{3}$ days.
2.	3	Son's 1 day's work = $\left(\frac{1}{3}-\frac{1}{5}\right)=\frac{2}{15}$ $\therefore$ The son alone can do the work in $\frac{15}{2}=7\frac{1}{2}$ days.
3.	1	Let leak will empty it in x hours $\therefore \frac{1}{8}-\frac{1}{x}=\frac{1}{10}$ $\frac{1}{8}$ - $\frac{1}{10}=\frac{1}{x}$ $\frac{1}{40}=\frac{1}{x}$. x = 40 hours
4.	1	Let waste pipe can empty the full cistem in x min $\frac{1}{10}+\frac{1}{15}-\frac{1}{x}=\frac{1}{18}$, $\frac{1}{10}+\frac{1}{15}-\frac{1}{18}=\frac{1}{x}$ $\frac{18+12-10}{180}=\frac{1}{x}=\frac{20}{180}=\frac{1}{x}$ x = 9 min
5.	2	2 (A+B+C)'s one day work = $\frac{1}{3}+\frac{1}{4}+\frac{1}{6}=\frac{9}{12}=\frac{3}{4}$ $\therefore$ (A + B + C)'s one day work= $\frac{3}{4}\times\frac{1}{2}=\frac{3}{8}$ $\therefore$ They will do the work in $\frac{8}{3}$ days or $2\frac{2}{3}$ days.
6.	1	Working together, they will finish the work in $\frac{1}{20}+\frac{1}{30}=\frac{1}{12}$ = 12 days.
7.	2	One minute work of three pipes = $\frac{1}{32}+\frac{1}{36}-\frac{1}{20}=\frac{45+40-72}{1440}=\frac{13}{1440}$ $\therefore$ They will fill the full tank in = $\frac{1440}{13}$ min They will fill $\frac{1}{3}$ of the tank = $\frac{1440}{13}\times\frac{1}{3}=\frac{480}{13}$ min = $36\frac{12}{13}$ min .
8.	4	A's 1 day work = $\frac{1}{15}$; B's 1 day work = $\frac{1}{20}$, (A + B)'s 1 day's work = $\left(\frac{1}{15}+\frac{1}{20}\right)=\frac{7}{60}$ (A + B)'s 4 day's work = $\left(\frac{7}{60}\times 4\right)=\frac{7}{15}$ $\therefore$ Remaining work = $\left(1-\frac{7}{15}\right)=\frac{8}{15}$
9.	4	If both taps are opened, cistem will be filled $\frac{1}{12\left(\frac{1}{4}-\frac{1}{6}\right)}$ parts in one minute, therefore; full cistern will be filled in 12 minutes.
10.	1	If both pipes are working, then the tank would be filled in $\frac{1}{20}-\frac{1}{60}=\frac{1}{30}$ = 30 min.